



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

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Student Names:

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Course Name: Deep Learning

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Title: CNN-Based Traffic Sign Recognition System

Assignment Description:

This assignment involved designing and implementing a CNN-based Traffic Sign Recognition System covering the full deep-learning workflow — data preparation, model design, training, evaluation and analysis. We chose a compact 3-block CNN (Conv→BatchNorm→ReLU→MaxPool ×3, ~685K parameters) over a deeper network to keep training fast and inference real-time-capable while still reaching high accuracy. Batch normalization and dropout(0.4) were added specifically to stabilize training and control overfitting given a moderate dataset size. Since a licensed real-world dataset wasn't available within the project timeline, we built a procedural synthetic-image generator with realistic augmentation (rotation, occlusion, motion blur, brightness jitter, noise) so the same pipeline generalizes directly to a real benchmark like GTSRB. The main challenge was avoiding an unrealistically easy dataset — an early version of the generator produced 100% accuracy with no meaningful errors to analyze. We addressed this by adding visually similar sign classes (two speed-limit variants) and heavier augmentation, which produced genuine, explainable misclassifications rather than trivial results. Balancing training stability (via BatchNorm/learning-rate scheduling) against overfitting was another recurring trade-off during tuning. Beyond CNN theory, we learned how augmentation design directly shapes both a model's robustness and its failure modes, how to read a confusion matrix diagnostically rather than just checking overall accuracy, and how architectural choices (kernel size, pooling, dropout) trade off accuracy against training/inference cost in a way that matters for real deployment. The project maps to **SDG 9 (Industry, Innovation and Infrastructure)** and **SDG 11 (Sustainable Cities and Communities)** by supporting safer, more automated intelligent-transportation and driver-assistance systems that can reduce accidents and improve traffic infrastructure efficiency.

- **CO1** (learning algorithms, backpropagation) — attained through implementing and training the CNN via gradient-based optimization (Adam) and cross-entropy loss.
- **CO2** (designing deep architectures) — attained through the CNN architecture design, layer configuration and justification in the design section.
- **CO3** (developing and tuning CNNs for vision tasks) — attained through model tuning, augmentation strategy, evaluation (accuracy/precision/recall/F1/confusion matrix) and the trade-off analysis of the finished system.

Guide Signature

Student Signature